

Notes: Love (RFS, 2003)

Financial Development and Financing Constraints: International Evidence from the Structural Investment Model

Contents

1	Big picture	2
2	Incremental contribution of the paper	2
2.1	Contribution relative to finance-growth evidence	2
2.2	Contribution relative to firm-level investment literature	2
2.3	Contribution relative to legal and institutional explanations	3
3	Data and measurement	3
3.1	Firm-level sample	3
3.2	Firm-level variables	3
3.3	Country-level financial development variables	4
3.4	Legal and institutional variables	4
4	Model and empirical specification	4
4.1	Firm value maximization problem	4
4.2	Euler equation	5
4.3	Financing-constraints specification	5
4.4	Marginal product of capital and adjustment costs	5
4.5	Linear empirical model	6
4.6	Estimation method	6
5	Identification and interpretation	6
5.1	Identifying financing constraints	6
5.2	Identifying the effect of financial development	6
5.3	Potential threats and robustness	6
6	Tables and figures	7
6.1	Tables 1 and 2: variable definitions and sample coverage	7
6.2	Tables 3 and 4: summary statistics and correlations	7
6.3	Tables 5 and 6: main Euler equation results	8
6.4	Tables 7 and 8: robustness tests	9
6.5	Figure 1: country-level cash coefficients and financial development	10
7	Short summary	11
8	Conclusion	11
8.1	Core conclusion	11
8.2	Economic intuition	11
8.3	Takeaway	11

1 Big picture

- **Question.** Does financial development reduce firm-level financing constraints, thereby improving the allocation of investment across firms and over time?
- **Main empirical strategy.** The paper estimates structural investment Euler equations using firm-level panel data across many countries, then tests whether the sensitivity of firms' effective discount factors to internal liquidity is smaller in financially developed countries.
- **Core mechanism.** Financing frictions make internal funds valuable. When a firm has low liquidity, the shadow value of external finance rises, the effective discount factor falls, and the firm postpones investment. Better financial markets reduce this wedge.
- **Main variable.** Financial development, FD_c , is a country-level index constructed from stock market development and financial intermediary development.
- **Key coefficient.** The central test is the interaction between lagged cash stock and financial development, $Cash_{i,t-1} \times FD_c$, in the investment Euler equation.
- **Main result.** The cash coefficient is positive and the cash-by-financial-development interaction is negative. Internal liquidity matters for investment timing, but it matters less in countries with more developed financial systems.
- **Economic magnitude.** The implied cost-of-capital effect from cash holdings is about twice as large in low-financial-development countries as in countries with average financial development.
- **Heterogeneity.** Small firms are disproportionately more constrained in low-financial-development environments, consistent with financial development relaxing constraints especially for firms with less access to capital markets.
- **Legal institutions.** Strong legal systems, low corruption, lower expropriation risk, and better accounting standards are associated with weaker financing constraints, but the financial-development interaction remains important.
- **Takeaway.** Financial development affects growth not only through aggregate correlations but through a microeconomic channel: it reduces distortions in firms' intertemporal investment decisions.

2 Incremental contribution of the paper

2.1 Contribution relative to finance-growth evidence

- Earlier cross-country finance-growth papers show that financial development predicts economic growth and productivity growth.
- This paper adds a firm-level mechanism: financial development reduces financing constraints that distort investment decisions.
- Instead of simply correlating finance with growth, the paper shows how financial development affects the firm's stochastic discount factor and hence investment timing.

Interpretation: The incremental contribution is a micro-foundation for the finance-growth relationship. If financial development reduces the wedge between internal and external finance, investment can move toward firms with high marginal returns, improving aggregate capital allocation.

2.2 Contribution relative to firm-level investment literature

- The U.S.-focused investment literature estimates Euler equations or Q models to detect financing constraints.

- Love extends the Euler-equation method to a large international panel, exploiting cross-country institutional and financial-market variation.
- The paper avoids relying on marginal q , which is difficult to measure internationally because stock-market valuations may be less informative in low-development markets.

Interpretation: The paper’s identification is not based only on within-country firm liquidity. It also asks whether the liquidity-investment wedge systematically varies with financial development.

2.3 Contribution relative to legal and institutional explanations

- The law-and-finance literature emphasizes legal origin, shareholder protection, corruption, accounting standards, and expropriation risk.
- Love tests whether these legal-institutional measures explain financing constraints directly and whether they subsume financial development.
- The results show that legal institutions matter, but the financial-development interaction remains significant in several specifications.

Interpretation: The paper helps separate financial development as a market-depth/intermediation concept from legal institutional quality, while recognizing that they are correlated.

3 Data and measurement

3.1 Firm-level sample

- The firm-level data come from Worldscope.
- The sample contains large publicly traded firms across 36 countries over 1988–1998.
- The final sample contains 48,132 firm-year observations and 4,794 firms.
- The number of firms varies substantially across countries; the United States and United Kingdom have the most firms, while several emerging economies have much smaller samples.

Interpretation: The sample is tilted toward large public firms. This makes the results conservative in one sense: if large listed firms are constrained in financially underdeveloped countries, smaller private firms are likely even more constrained.

3.2 Firm-level variables

The key firm-level variables are:

- $PPENT$: property, plant, and equipment net of depreciation;
- $CAPEX$: capital expenditure;
- DA : depreciation and amortization;
- $K = PPENT - CAPEX + DA$: beginning-of-period capital;
- $I/K = CAPEX/K$: investment-to-capital ratio;
- S/K : sales-to-capital ratio, used as a proxy for the marginal product of capital;
- $Cash$: cash plus equivalents scaled by total assets or capital;
- CF : cash flow;

- *COGS*: cost of goods sold;
- *Size*: log total assets in U.S. dollars.

Interpretation: The investment Euler equation relies on capital, investment, sales, and cash. Sales-to-capital proxies for marginal profitability, while cash captures internal liquidity and financing constraints.

3.3 Country-level financial development variables

Financial development is constructed from two broad components:

- **Stock market development** (*STKMKT*): standardized measures of market capitalization/GDP, total value traded/GDP, and turnover.
- **Financial intermediary development** (*FININT*): standardized measures of bank and nonbank intermediary liabilities/GDP and private credit/GDP.

The composite index is

$$FD_c = STKMKT_c + FININT_c. \quad (1)$$

Interpretation: The *FD* index captures both market-based and bank/intermediary-based financial development. This matters because external finance can come through either securities markets or financial intermediaries.

3.4 Legal and institutional variables

The paper also considers:

- legal origin;
- efficiency of legal system and rule of law;
- corruption;
- risk of expropriation;
- accounting standards;
- GDP per capita and GDP growth.

Interpretation: These variables are used to test whether the effect attributed to financial development is really an effect of broader legal and institutional quality.

4 Model and empirical specification

4.1 Firm value maximization problem

The firm maximizes the expected discounted value of dividends:

$$V_t(K_t, \xi_t) = \max_{\{I_{t+s}\}_{s \geq 0}} \left[D_t + E_t \sum_{s=1}^{\infty} \beta_{t+s-1} D_{t+s} \right]. \quad (2)$$

Dividends satisfy the sources-equal-uses constraint:

$$D_t = \Pi(K_t, \xi_t) - C(I_t, K_t) - I_t. \quad (3)$$

Capital evolves according to

$$K_{t+1} = (1 - \delta)K_t + I_t. \quad (4)$$

The financing constraint is represented as a nonnegative-dividend constraint:

$$D_t > 0. \quad (5)$$

Interpretation: A binding nonnegative-dividend constraint means the firm cannot freely raise external equity. The multiplier on this constraint is the shadow cost of external finance.

4.2 Euler equation

The investment Euler equation equates the marginal cost of investing today with the discounted marginal benefit of postponing investment:

$$1 + \left(\frac{\partial C}{\partial I} \right)_t = \beta_t E_t \left[\Theta_t \left\{ \left(\frac{\partial \Pi}{\partial K} \right)_{t+1} + (1 - \delta) \left(1 + \left(\frac{\partial C}{\partial I} \right)_{t+1} \right) \right\} \right]. \quad (6)$$

The financial-frictions component of the discount factor is

$$\Theta_t = \frac{1 + \lambda_{t+1}}{1 + \lambda_t}, \quad (7)$$

where λ_t is the shadow value of the nonnegative-dividend constraint.

Interpretation: If external finance is costly today relative to tomorrow, then $\lambda_t > \lambda_{t+1}$, the effective discount factor falls, and the firm delays investment.

4.3 Financing-constraints specification

The paper parameterizes the financing-frictions component as

$$\Theta_{it} = a_{0i} + (a_1 + a_2 F D_c) Cash_{i,t-1}. \quad (8)$$

The hypothesis is:

$$a_1 > 0, \quad a_2 < 0. \quad (9)$$

Interpretation: Cash should relax financing constraints, so $a_1 > 0$. Financial development should make internal cash less important, so $a_2 < 0$.

4.4 Marginal product of capital and adjustment costs

The marginal product of capital is proxied by sales-to-capital:

$$MPK_{it} = \theta_i \left(\frac{S}{K} \right)_{it} \approx const. + \bar{\theta} \left(\frac{S}{K} \right)_{it}. \quad (10)$$

The marginal adjustment cost is assumed linear:

$$\left(\frac{\partial C}{\partial I} \right)_{it} = \left(\frac{I}{K} \right)_{it} - g \left(\frac{I}{K} \right)_{i,t-1} - \nu_i. \quad (11)$$

Interpretation: Lagged investment enters to capture persistence in investment-to-capital ratios.

4.5 Linear empirical model

After substitution, linearization, and replacing expectations with realizations plus rational-expectations errors, the empirical specification is:

$$\left(\frac{I}{K}\right)_{it} = \beta_1 \left(\frac{I}{K}\right)_{i,t+1} + \beta_2 \left(\frac{I}{K}\right)_{i,t-1} + \beta_3 \left(\frac{S}{K}\right)_{it} \quad (12)$$

$$+ \beta_4 Cash_{i,t-1} + \beta_5 Cash_{i,t-1} FD_c + f_i + d_{c,t} + \varepsilon_{it}. \quad (13)$$

The key predictions are:

$$\beta_4 > 0, \quad \beta_5 < 0. \quad (14)$$

Interpretation: The cash term measures financing constraints. The cash-by-financial-development term asks whether these constraints are weaker in financially developed economies.

4.6 Estimation method

- Firm fixed effects are removed using forward mean differencing, following Arellano and Bover.
- Country-time effects are removed by country-time differencing.
- The model is estimated by GMM.
- Instruments include first and second lags of investment, sales, cash, cash flow, cost of goods sold, industry dummies, and interactions with financial development and other variables.
- Overidentification tests and bootstrapped tests are used to assess model validity.

Interpretation: The estimation is designed to respect the dynamic panel structure and the predetermined nature of investment and cash variables.

5 Identification and interpretation

5.1 Identifying financing constraints

The paper identifies financing constraints from the role of internal liquidity in the Euler equation. If capital markets are perfect, cash should not enter the investment Euler equation once future profitability, adjustment costs, firm effects, and country-time shocks are controlled for.

Interpretation: A significant cash coefficient implies that internal funds affect the intertemporal allocation of investment, consistent with financing constraints.

5.2 Identifying the effect of financial development

The cross-country interaction $Cash_{i,t-1} \times FD_c$ tests whether internal liquidity matters less in financially developed countries.

Interpretation: This is a difference-in-slopes design: the cash-investment Euler equation sensitivity is compared across countries with different financial development levels.

5.3 Potential threats and robustness

- **Size effect:** small firms are more likely constrained; the paper interacts cash with size and with size times financial development.

- **Business cycles:** the paper interacts cash with country-level GDP growth to test whether booms reduce financing constraints.
- **Legal system:** the paper replaces and supplements financial development with legal-institutional variables.
- **Country-specific coefficients:** the paper estimates country-level cash coefficients and correlates them with financial development.

Interpretation: These robustness exercises test whether financial development is capturing firm size, macro cycles, or legal institutions. The main financial-development result survives.

6 Tables and figures

6.1 Tables 1 and 2: variable definitions and sample coverage

Read:

- Table 1 defines firm-level and country-level variables.
- Table 2 shows that the sample covers 36 countries and 48,132 observations.
- The United States and United Kingdom are the largest country samples, while many emerging markets have fewer observations.

Economic message: The paper's strength comes from combining firm-level investment data with cross-country variation in financial development.

Table 1
Variable definitions

Abbreviation	Description
<i>Firm-level variables (from Worldscope)</i>	
PPENT	Property plant and equipment, net of depreciation
CAPEX	Capital expenditure
DA	Depreciation and amortization expense
K	Beginning period capital = PPENT – CAPEX + DA
IK, I/K	Investment to capital ratio = CAPEX/K
SK, S/K	Sales to capital ratio = sales/K
Cash	Cash plus equivalents scaled by total assets (or scaled by K for robustness checks)
CF	Cash flow (net income + DA), scaled by K
COGS	Cost of goods sold, scaled by K
Size	Log of total assets in U.S. dollars
Rank	Ranking based on size of PPENT (first, ranked by year, then averaged over the years), largest firm in each country has rank equal to one.
Weight	Weight is a country-level variable equal to one over the number of valid observations per country.
Industry dummies	For manufacturing industries the dummies are on a two digit SIC level and for the rest of industries they are on a one digit level.
<i>Country-level variables</i>	
STKMKT	Stock market development is Index1 from Demurgue-Kunt and Levine (1996), equals to the sum of (standardized indices of) market capitalization to GDP, total value traded to GDP, and turnover (total value traded to market capitalization).
FININT	Financial intermediary development is Index1 from Demurgue-Kunt and Levine (1996), equals to the sum of (standardized indices of) ratio of liquid liabilities to GDP, and ratio of domestic credit to private sector to GDP.
FD	Financial development = STKMKT + FININT.
Legal origin	Country's legal origin categorized into four groups: English, French, German or Scandinavian, from La Porta et al. (1998).
Efficiency, rule of law	Efficiency of legal system and rule of law are two measure of the quality of law enforcement, from La Porta et al. (1998).
Expropriation	Risk of expropriation is the risk of outright confiscation or forced confiscation by the government, from La Porta et al. (1998).
Corruption	The measure of corruption, from La Porta et al. (1998).
GNP PC	Log of GNP per capita in U.S. dollars in 1994, World Development Report 1996.
grGDP	Annual real growth rate of GDP, IFS.

Table 1: Variable definitions

Table 2
Sample coverage across countries

Country	Country code	Number of observations	Percent of total observations	Number of firms
Argentina	AR	177	0.003	21
Austria	AT	424	0.008	45
Australia	AU	1359	0.026	135
Belgium	BE	474	0.009	50
Brazil	BR	593	0.012	68
Canada	CA	2950	0.057	283
Switzerland	CH	941	0.017	102
Chile	CL	390	0.007	46
Colombia	CO	108	0.003	10
Germany	DE	3688	0.067	379
Denmark	DK	927	0.018	97
Spain	ES	838	0.016	93
Finland	FI	656	0.013	65
France	FR	2794	0.055	296
United Kingdom	GB	8038	0.166	801
Indonesia	ID	464	0.009	59
Israel	IL	125	0.003	17
India	IN	1353	0.025	131
Italy	IT	1049	0.019	104
Japan	JP	3447	0.078	291
South Korea	KR	903	0.021	60
Mexico	MX	455	0.008	55
Malaysia	MY	1216	0.025	121
Netherlands	NL	1142	0.021	117
Norway	NO	607	0.011	59
New Zealand	NZ	280	0.005	28
Philippines	PH	222	0.005	20
Pakistan	PK	385	0.007	48
Portugal	PT	206	0.004	19
Sweden	SE	977	0.019	92
Singapore	SG	617	0.014	61
Thailand	TH	855	0.018	106
Turkey	TR	97	0.002	6
USA	US	8293	0.175	800
Venezuela	VE	74	0.001	8
South Africa	ZA	1008	0.019	101
Total		48,132		4794
Average number of firms per country				133
Average number of firms per country, excluding U.S. and U.K.				94
Median number of firms per country, excluding U.S. and U.K.				80

Table 2: Sample coverage across countries

6.2 Tables 3 and 4: summary statistics and correlations

Read:

- Table 3 reports country-level means and medians for cash, investment-to-capital, sales-to-capital, and financial development measures.
- Financial development varies sharply across countries, with the United States, United Kingdom, Japan, and Switzerland high on the index.
- Table 4 reports correlations among firm variables and country-level institutional variables.
- Financial development is correlated with legal and institutional quality, motivating later robustness tests.

Economic message: There is enough cross-country variation in both firm liquidity and financial development to test whether financing constraints vary systematically with financial-market development.

Table 3
Descriptive statistics for key variables

Country	Cash		I/K		S/K		Financial development		
	Mean	Median	Mean	Median	Mean	Median	FD	FININT	STKMKT
Argentina	0.08	0.04	0.19	0.13	1.6	1.2	-1.38	-0.79	-0.59
Austria	0.10	0.07	0.25	0.20	4.4	3.0	-0.27	-0.12	-0.15
Australia	0.08	0.05	0.26	0.18	3.3	2.3	0.42	0.23	0.19
Belgium	0.10	0.08	0.25	0.20	4.0	3.7	-0.82	-0.35	-0.47
Brazil	0.08	0.04	0.12	0.09	1.7	1.0	-1.04	-0.75	-0.29
Canada	0.07	0.02	0.23	0.18	3.1	1.5	0.03	-0.06	0.09
Switzerland	0.14	0.11	0.23	0.15	3.9	2.6	2.2	1.45	0.75
Chile	0.07	0.04	0.21	0.16	1.6	1.3	-0.75	-0.29	-0.46
Colombia	0.08	0.04	0.26	0.14	3.6	1.9	-1.6	-0.72	-0.88
Germany	0.08	0.05	0.31	0.25	5.5	4.8	1.68	0.3	1.38
Denmark	0.15	0.14	0.24	0.21	4.4	3.7	-0.49	-0.12	-0.37
Spain	0.06	0.04	0.14	0.09	2.8	1.8	-0.14	0.11	-0.25
Finland	0.09	0.08	0.38	0.21	5.0	3.2	-0.41	0.12	-0.53
France	0.12	0.09	0.27	0.21	6.5	6.0	0.1	0.31	-0.21
UK	0.09	0.06	0.22	0.17	4.6	4.0	1.68	0.45	1.23
Indonesia	0.14	0.10	0.37	0.23	3.9	2.6	-1.17	-0.46	-0.71
Israel	0.11	0.10	0.30	0.24	3.7	3.0	0.01	-0.07	0.08
India	0.04	0.03	0.27	0.19	3.5	2.6	-0.7	-0.44	-0.26
Italy	0.12	0.09	0.26	0.17	4.4	3.0	-0.64	-0.13	-0.51
Japan	<i>0.19</i>	<i>0.17</i>	0.22	0.19	4.1	3.3	<i>3.3</i>	1.31	2.02
South Korea	0.08	0.06	0.31	0.23	3.9	3.0	0.84	-0.21	1.05
Mexico	0.08	0.06	0.11	0.10	1.6	1.3	-0.85	-0.71	-0.14
Malaysia	0.08	0.05	0.23	0.16	2.7	1.8	1.19	0.29	0.9
Netherlands	0.10	0.05	0.24	0.20	5.1	3.8	0.66	0.34	0.32
Norway	0.14	0.12	0.33	0.22	3.6	2.1	-0.15	0.03	-0.18
New Zealand	0.04	0.02	0.17	0.13	3.3	2.8	-0.53	-0.2	-0.33
Philippines	0.12	0.07	0.37	0.22	2.6	1.4	-1.15	-0.61	-0.54
Pakistan	0.11	0.04	0.26	0.19	4.8	2.5	-1.28	-0.46	-0.82
Portugal	0.06	0.03	0.22	0.13	3.3	2.0	-0.67	-0.06	-0.61
Sweden	0.12	0.08	0.31	0.19	4.7	3.8	-0.31	-0.21	-0.1
Singapore	<i>0.19</i>	0.15	0.28	0.22	3.6	2.5	1.6	0.56	1.04
Thailand	0.06	0.03	0.40	0.23	4.5	2.5	0.36	-0.02	0.38
Turkey	0.13	0.07	<i>0.56</i>	<i>0.50</i>	7.5	5.8	-1.2	-0.59	-0.61
U.S.	0.09	0.04	0.24	0.19	4.8	3.8	1.35	0.14	1.21
Venezuela	0.09	0.06	0.21	0.13	1.5	1.1	-1.26	-0.52	-0.74
South Africa	0.09	0.06	0.22	0.19	4.7	3.8	0.25	-0.23	0.48
Mean	0.10	0.07	0.26	0.19	3.82	2.80	-0.03	-0.07	0.04
Median	0.09	0.06	0.25	0.19	3.87	2.62	-0.29	-0.12	-0.20
Std	0.04	0.04	0.08	0.07	1.33	1.22	1.14	0.50	0.73

Summary statistics by country for main variables. Variables definitions are given in Table 1. Outliers (far away maximum or minimum) are underlined.

Table 3: Descriptive statistics by country

Table 4
Panel A. Cross-country correlations of firm level variables

	Country means				Country medians			
	FD	Cash	IK	SK	FD	Cash	IK	SK
FD		0.15 (0.35)	0.02 (0.87)	0.34 (0.04)		0.21 (0.22)	0.21 (0.22)	0.44 (0.006)
Cash	0.21 (0.23)		0.32 (0.04)	0.36 (0.02)	0.21 (0.22)		0.36 (0.02)	0.32 (0.04)
IK	0.09 (0.59)	0.27 (0.099)		0.55 (0.0002)	0.29 (0.096)	0.35 (0.03)		0.59 (0.0001)
SK	0.41 (0.015)	0.24 (0.14)	0.49 (0.002)		0.45 (0.008)	0.23 (0.15)	0.56 (0.0003)	

Panel B. Correlations of country-level institutional characteristics

	FD	FININT	STKMKT	Efficiency	Corruption	Exprop.	Accounting	GNPPC
FININT	0.90 (0)							
STKMKT	0.95 (0)	0.73 (0)						
Efficiency	0.52 (0.001)	0.52 (0.001)	0.45 (0.005)					
Corruption	0.51 (0.001)	0.55 (0.001)	0.40 (0.01)	0.83 (0)				
Expropriation	0.57 (0.001)	0.64 (0)	0.39 (0.02)	0.72 (0)	0.83 (0)			
Accounting	0.34 (0.05)	0.28 (0.09)	0.34 (0.04)	0.31 (0.06)	0.41 (0.01)	0.36 (0.03)		
GNP PC	0.56 (0)	0.61 (0.0)	0.46 (0.004)	0.74 (0)	0.87 (0)	0.84 (0)	0.46 (0.005)	
GDP US	0.54 (0)	0.41 (0.01)	0.49 (0.002)	0.20 (0.22)	0.26 (0.11)	0.46 (0.003)	0.15 (0.39)	0.42 (0.01)

Correlations of country-level means and medians of the firm level variables and country's institutional characteristics. Variables definitions are in Table 1. Panel A: below the diagonal are Pearson correlation coefficients, with two outlier countries excluded: JP (Japan) and TR (Turkey). Above the diagonal are Spearman correlations (robust to outliers) with all observations included. Including outliers for Pearson correlations results in significant correlation for FD and Cash (due to JP which is an outlier on both of these) and nonsignificant correlation for SK and FD (due to TR, which has very high SK and low FD). Panel B: Pearson correlations with all countries. (Excluding Japan makes correlations of GDP PC with FD and FININT significant at 6% and 2%, respectively; also correlation between FININT and log GDP becomes insignificant.) *P*-values are in parenthesis; bold are significant at 5% or better, underlined are significant at 10%.

Table 4: Correlations

6.3 Tables 5 and 6: main Euler equation results

Read:

- Table 5 shows a positive coefficient on lagged cash and a negative coefficient on cash times financial development.
- The interaction is negative and significant across the 25-largest, 50-largest, 150-largest, and all-weighted samples.
- The estimated cash effect is roughly twice as large in low financial development countries as in countries with average financial development.
- Table 6 splits countries into high-FD and low-FD samples.
- In the high-FD sample, cash coefficients are small and not significant; in the low-FD sample, cash coefficients are large and significant.

Economic message: These tables provide the core evidence: financial development reduces the importance of internal cash for investment timing.

Table 5
Main results on financial development and financing constraints

Model:	1 25 largest	2 50 largest	3 150 largest	4 All, weighted
IK_{it+1}	0.733 (0.105)***	0.643 (0.103)***	0.638 (0.093)***	0.353 (0.058)***
IK_{it-1}	0.231 (0.025)***	0.219 (0.017)***	0.218 (0.013)***	0.203 (0.008)***
SK_{it}	0.021 (0.008)***	0.017 (0.006)***	0.013 (0.005)***	0.033 (0.004)***
$Cash_{it-1}$	0.189 (0.063)***	0.143 (0.047)***	0.138 (0.039)***	0.218 (0.034)***
$Cash_{it-1} * FD_c$	-0.157 (0.039)***	-0.113 (0.032)***	-0.102 (0.027)***	-0.183 (0.024)***
Constant	0.000 (0.001)	-0.002 (0.001)**	-0.001 (0.001)	-0.002 (0.001)**
N observations	3533	6488	12474	21278
N firms	755	1436	2791	4794
p -value for J -statistic				
Chi-square	0.72	0.208	0.015	NA
Bootstrap	0.93	0.463	0.042	0.792

The dependent variable is IK_t , the model is given in Equation (10); variable definitions are in Table 1. The estimation is by GMM; country-time, and fixed effects are removed prior to estimation (see Section 2.1). Instruments are first and second lags of IK , SK , $Cash$, CFK , $COGS$, interactions of FD with IK , SK and $Cash$, and industry dummies. The firms are ranked based on the size of the capital stock. In the weighted regression, weights are equal to a value of one divided by the number of observations per country. P -values for J -statistic (test of overidentifying restrictions) are obtained using chi-square distribution or bootstrap simulation with 200 repetitions (the chi-square p -value is not available for weighted regressions). Heteroscedasticity adjusted standard errors in parentheses.

***, **, and * represent significance at 1%, 5%, and 10%, respectively.

Table 5: Main results

Table 6
Sample splits

Model:	High FD		Low FD	
	1 50 largest	2 150 largest	3 50 largest	4 150 largest
IK_{it+1}	0.629 (0.064)***	0.734 (0.094)***	0.766 (0.105)***	0.643 (0.126)***
IK_{it-1}	0.257 (0.022)***	0.231 (0.015)***	0.184 (0.024)***	0.181 (0.021)***
SK_{it}	0.014 (0.007)***	0.014 (0.005)***	0.008 (0.008)	0.019 (0.009)**
$Cash_{it-1}$	0.041 (0.044)	0.035 (0.036)	0.210 (0.072)***	0.258 (0.069)***
Constant	0.000 (0.002)	0.000 (0.001)	-0.001 (0.002)	-0.001 (0.002)
N observations	4335	9377	2579	3596
N firms	861	1962	686	969
p -value for J -statistic				
Chi-square	0.907	0.348	0.518	0.495
Bootstrap	0.983	0.528	0.792	0.593

The dependent variable is IK_t ; variable definitions are in Table 1. High FD and low FD are samples split on the median FD (reported in Table 3). The estimation is by GMM; country-time and fixed effects are removed prior to estimation (see Section 2.1). Instruments are first and second lags of IK , SK , $Cash$, CFK , $COGS$ and industry dummies. The firms are ranked based on the size of the capital stock. P -values for J -statistic (test of overidentifying restrictions) are obtained using chi-square distribution or bootstrap simulation with 200 repetitions. Heteroscedasticity adjusted standard errors in parentheses.

***, **, and * represent significance at 1%, 5%, and 10%, respectively.

Table 6: High- and low-financial-development samples

6.4 Tables 7 and 8: robustness tests

Read:

- Table 7 shows that smaller firms are more financially constrained and that the size effect is stronger in countries with low financial development.
- Business-cycle interactions do not eliminate the main financial-development interaction.
- Table 8 examines legal system indicators.
- Better legal systems, lower corruption, lower expropriation risk, and stronger accounting standards are associated with weaker financing constraints.
- Financial development remains statistically and economically important even when these legal indicators are included.

Economic message: The main finding is not only a size story, a business-cycle story, or a legal-origin story. Financial development has an independent relationship with financing constraints.

Table 7
Robustness: size effect and business cycles

Model:	Size Effect			Business Cycles	
	1	2	3	4	5
IK_{it+1}	0.534 (0.061)***	0.444 (0.057)**	0.437 (0.058)**	0.294 (0.071)**	0.215 (0.063)***
IK_{it-1}	0.197 (0.008)***	0.200 (0.008)***	0.199 (0.008)***	0.189 (0.009)***	0.194 (0.008)***
SK_{it}	0.026 (0.004)***	0.029 (0.004)***	0.029 (0.004)***	0.035 (0.004)***	0.037 (0.004)***
$Cash_{it-1}$	0.443 (0.102)***	0.493 (0.107)***	0.617 (0.119)***	0.226 (0.041)***	0.254 (0.042)
$Cash_{it-1} \times Size_i$	-0.057 (0.017)***	-0.059 (0.018)***	-0.083 (0.021)***		
$Cash_{it-1} \times FD_c$		-0.172 (0.024)***	-0.588 (0.112)**		-0.164 (0.025)***
$Cash_{it-1} \times Size_i \times FD_c$			0.078 (0.019)***		
$Cash_{it-1} \times grGDP_{it}$				-1.420 (0.622)**	-0.973 (0.632)*
Constant	-0.002 (0.001)***	-0.002 (0.001)***	-0.002 (0.001)***	-0.002 (0.001)***	-0.003 (0.001)***
N observations	21278	21278	21278	21009	21009
N firms	4794	4794	4794	4768	4768
Bootstrapped p -value for J -statistic:	0.142	0.193	0.228	0.973	0.993

The dependent variable is IK_t . The size is equal to the (log of) total assets in U.S. dollars, $grGDP$ is country-year real GDP growth rate. The estimation is by GMM; country-time and fixed effects are removed prior to estimation (see Section 2.1). Instruments are first and second lags of IK , SK , $Cash$, CF , $COGS$, size (GDP growth) and its interactions with $Cash$, IK and SK , interactions of FD with IK , SK , $Cash$, and size, and industry dummies. All the regressions are weighted regressions, weights are equal to a value of one divided by the number of observations per country. P -values for J -statistic (test of overidentifying restrictions) are obtained using bootstrap simulation with 200 repetitions. Heteroscedasticity adjusted standard errors in parentheses.

***, ** and * represent significance at 1%, 5%, and 10% level

Table 7: Size effects and business cycles

Table 8
Legal system indicators and financing constraints

Model:	Indicator:	$Cash_{it-1}$	$Cash_{it-1} \times Indicator_c$	$Cash_{it-1} \times FD_c$	Bootstrapped p -value for J -statistic:
1	Efficiency	0.685*** (0.101)	-0.062*** (0.012)		0.792
2	Efficiency	0.392*** (0.122)	-0.022 (0.016)	-0.153*** (0.030)	0.853
3	Rule of law	0.712*** (0.100)	-0.065*** (0.012)		0.738
4	Rule of law	0.564*** (0.112)	-0.044*** (0.014)	-0.139*** (0.026)	0.718
5	Corruption	0.716*** (0.098)	-0.069*** (0.012)		0.818
6	Corruption	0.507*** (0.117)	-0.039*** (0.015)	-0.133*** (0.029)	0.802
7	Expropriation	1.510*** (0.175)	-0.149*** (0.019)		0.643
8	Expropriation	1.210*** (0.211)	-0.112*** (0.024)	-0.111*** (0.027)	0.463
9	Accounting	1.140*** (0.158)	-0.015*** (0.002)		0.302
10	Accounting	0.761*** (0.193)	-0.009*** (0.003)	-0.126*** (0.027)	0.387
11	English origin	0.263*** (0.039)	-0.223*** (0.061)		0.313
12	English origin	0.245*** (0.039)	-0.064 (0.066)	-0.169*** (0.025)	0.568

The dependent variable is IK_t , the "baseline" model is given in Equation (10) with FD interactions replaced or supplemented with each of the indicator variable interactions (the rest of coefficients are not reported). Variable definitions are in Table 1. The estimation is by GMM, country-time and fixed effects are removed prior to estimation (see Section 2.1). Instruments are first and second lags of IK , SK , $Cash$, CF , $COGS$, interactions of FD and appropriate indicator with IK , SK and $Cash$, and industry dummies. All the regressions are weighted regressions, weights are equal to a value of one divided by the number of observations per country. All regressions include 21,278 observations (4794 firms) except for models with accounting standards which include 21,032 observations (4687 firms). P -values for J -statistic (test of overidentifying restrictions) are obtained using bootstrap simulation with 200 repetitions.

Table 8: Legal system indicators

6.5 Figure 1: country-level cash coefficients and financial development

Read:

- Figure 1 plots first-stage country-level cash coefficients against financial development.
- The fitted line slopes downward.
- The cash coefficient is larger in financially underdeveloped countries and smaller in more financially developed countries.
- The relationship is statistically significant even when influential observations are removed.

Economic message: The figure gives a transparent visual version of the paper's central claim: financially developed countries exhibit weaker investment-cash sensitivity in the Euler equation.

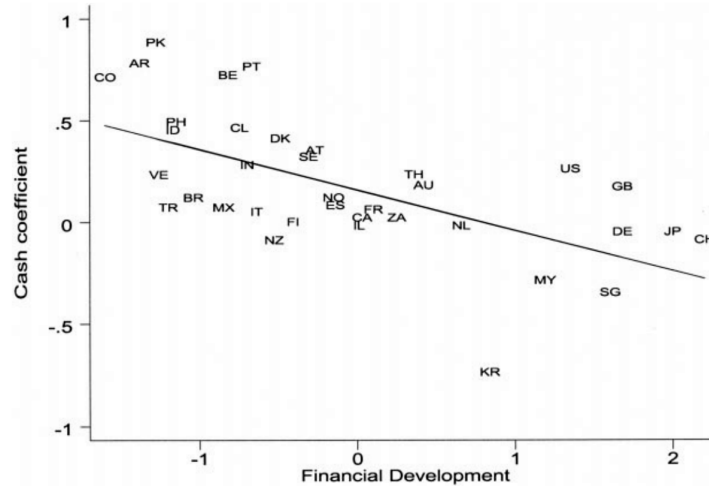


Figure 1
Cash coefficients and financial development

Regression line: -0.18, significant at 1% (the standard error is 0.05 and R^2 is 0.35). Without KR: -0.16,

7 Short summary

- The paper estimates investment Euler equations using firm-level data from 36 countries.
- Financing constraints are modeled as the effect of cash on the firm's stochastic discount factor.
- The main prediction is that the cash coefficient is positive and the cash-by-financial-development interaction is negative.
- Empirically, lagged cash matters for investment timing, but it matters less in financially developed countries.
- Low-financial-development countries show strong cash effects; high-financial-development countries do not.
- Small firms are especially constrained in low-financial-development countries.
- Legal and institutional variables matter but do not fully absorb the financial-development effect.
- The paper provides firm-level evidence that financial development improves capital allocation by reducing financing constraints.

8 Conclusion

8.1 Core conclusion

Love shows that financial development reduces firm-level financing constraints. In countries with less developed financial systems, firms' investment timing depends more strongly on internal cash holdings, implying a higher wedge between internal and external finance. In countries with more developed financial markets and intermediaries, this liquidity wedge is weaker.

8.2 Economic intuition

When external finance is costly, a firm with low internal cash postpones investment even if investment opportunities are good. This distortion lowers the efficiency of intertemporal capital allocation. Financial development reduces the wedge by improving access to outside funds and lowering the cost of external finance.

8.3 Takeaway

The paper connects two literatures: finance and growth, and firm-level financing constraints. Its key lesson is that financial development matters because it changes real investment behavior at the firm level, not only because it correlates with aggregate growth outcomes.